

Invitation to the
public doctoral defence of

Ruben Janssens

Visually Grounded Adaptive Social
Dialogue for Robots

Adaptieve sociale dialoog voor robots
verankerd in een visuele context

Positioning of the dissertation

For decades, researchers have attempted to build social robots that can fluently engage in conversation. While we know that robots offer great benefits, e.g. in education or care contexts, they repeatedly ran into the complexity of conversation.

One of those complex aspects is how in conversations, we use more than only words. We also adapt to what we see, such as when we make references to our surroundings or to our conversation partner, or when we express feedback about how the conversation is going through non-verbal communication.

This doctoral dissertation investigated how robots can adapt to that visual information in conversations. To achieve this aim, it used, for the first time, foundation vision-language models, which also lay at the basis of popular AI applications such as ChatGPT. This technology allows robots to always provide a coherent response, even when the user says something unexpected.

This research showed that this also works for visual information: systems that leverage vision-language models can accurately refer to a wide range of relevant visual cues. At the same time, these models still lack certain social interaction skills, such as knowing what is appropriate to mention at what time, and planning forward in a conversation.

In conclusion, this doctoral dissertation showed that foundation models allow social robots' conversational abilities to take a leap forward, but also that modelling social interactions remains a great, interdisciplinary, challenge for science and engineering. Nevertheless, this challenge holds the potential to lead to impactful new technologies and insights about the human mind.

Curriculum Vitae



Ruben Janssens was born on August 2nd, 1998, in Brasschaat, Belgium. In 2021, he obtained the degree of Master of Science in Computer Science Engineering from Ghent University.

In September 2021, he joined the AI and Robotics research team (AIRO) at IDLab, within the department Electronics and Information Systems (ELIS), as a PhD researcher. This research was funded by the Flanders Artificial Intelligence Research Programme. Ruben also spent four months as a visiting researcher at KTH Royal Institute of Technology in Stockholm, Sweden, hosted by Prof Gabriel Skantze, funded by an FWO travel grant.

His doctoral research consisted of developing artificial intelligence that allow social robots to engage in fluent conversations with people, adapting to visual input during those conversations. This research resulted in one article published in a scientific journal, twelve contributions to scientific conferences, of which eight as first author, and one chapter in a scientific book. Three manuscripts are currently still under review.

Besides scientific publications, Ruben also supported our faculty's educational efforts, assisting in courses, projects, and master's theses, and our university's policy and governance. He also contributed to the scientific community, as co-organiser of FEARS 2022 and main organiser of the 2022 HRI Winter School in Ghent and HRI Pioneers 2025 in Melbourne, Australia. Finally, he was also committed to communicating about scientific research to the general public, through various demos, workshops, and presentations.

You are warmly invited to the public
doctoral defence of

Ruben Janssens

This defence will take place on
Tuesday March 31st, 2026, at 17h30

in

Auditorium Jozef Plateau,
Plateau-Rozier building,
Jozef Plateaustraat 22,
9000 Gent

The defence will take place in Dutch,
with live subtitling in English.

After the defence, you are also warmly invited to a
reception with food and drinks in the same building.

Please confirm your attendance before Monday
March 23rd, through
<https://forms.cloud.microsoft/e/2HmQYv7Vs8>.

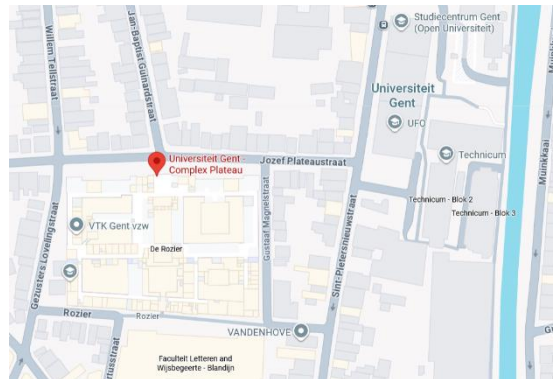
The defence will also be livestreamed, accessible
through the following Microsoft Teams link:
<https://teams.microsoft.com/meet/36314987331704?p=qGHPSQGc2Jx7f0JTmJ>.

Location

Auditorium Jozef Plateau (Auditorium P)
Plateau-Rozier building (ground floor)
Jozef Plateaustraat 22
9000 Gent

The building is easily accessible on foot, by bicycle, and
through public transport (stops Bagattenstraat or Zuid;
due to road works, no tram currently stops close by).
Parking is possible in the car parks Savaanstraat, Zuid,
and Sint-Pietersplein.

After entering the building, you walk through the glass
double doors on your left. The auditorium can be found
at the end of the corridor, on your left-hand side.



Advisors

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